Motor Full-Load Current Chart

Full-load current by horsepower for three-phase and single-phase motors, as used to size motor circuit conductors and protection.

For sizing a motor circuit, the full-load current comes from the code table for the motor's horsepower and voltage, not from its nameplate. A 50 hp three-phase motor is taken as 65 A at 460 V and 130 A at 230 V. The table figure is used for the conductors, the short-circuit protection and the disconnect; the nameplate current is used only for the overload protection.

Three-phase induction motors

hp	208 V	230 V	460 V	575 V
1/2	2.4	2.2	1.1	0.9
3/4	3.5	3.2	1.6	1.3
1	4.6	4.2	2.1	1.7
1-1/2	6.6	6	3	2.4
2	7.5	6.8	3.4	2.7
3	10.6	9.6	4.8	3.9
5	16.7	15.2	7.6	6.1
7-1/2	24.2	22	11	9
10	30.8	28	14	11
15	46.2	42	21	17
20	59.4	54	27	22
25	74.8	68	34	27
30	88	80	40	32
40	114	104	52	41
50	143	130	65	52
60	169	154	77	62
75	211	192	96	77
100	273	248	124	99
125	343	312	156	125
150	396	360	180	144
200	528	480	240	192
250	—	—	302	242
300	—	—	361	289
350	—	—	414	336
400	—	—	477	382
450	—	—	515	412
500	—	—	590	472

Full-load current in amperes, for squirrel-cage and wound-rotor induction motors running at usual speeds with normal torque characteristics. A dash means that size is not listed at that voltage.

Single-phase motors

hp	115 V	230 V
1/6	4.4	2.2
1/4	5.8	2.9
1/3	7.2	3.6
1/2	9.8	4.9
3/4	13.8	6.9
1	16	8
1-1/2	20	10
2	24	12
3	34	17
5	56	28
7-1/2	80	40
10	100	50

Full-load current in amperes.

Where the values come from

The three-phase currents correspond to NEC Table 430.250 and the single-phase currents to Table 430.248. They are reproduced here for reference and are not computed by Voltfield. The figures are the same pattern throughout: the 230 V current is twice the 460 V current, and the 208 V current is ten percent higher than the 230 V one. Check any value you intend to rely on against the edition of the code your jurisdiction has adopted.

These are code values, not Voltfield's. NEC and National Electrical Code are trademarks of the National Fire Protection Association, which publishes the code. Voltfield is not affiliated with it. The published code, in the edition adopted where you are, is the authority.

What this chart does not tell you

These are code values for sizing, and they are deliberately conservative. An actual motor, particularly a high-efficiency one, usually draws less. The tables do not cover low-speed motors of 1,200 rpm or less, high-torque motors or multispeed motors, which can draw more and are sized from the nameplate. They do not give locked-rotor current, and they do not size the overload device.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/motor-full-load-current-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.